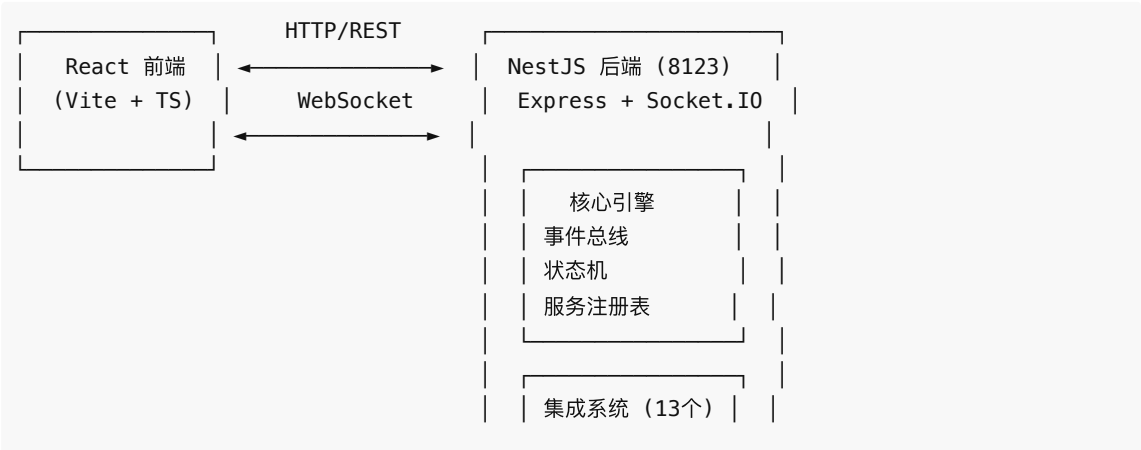


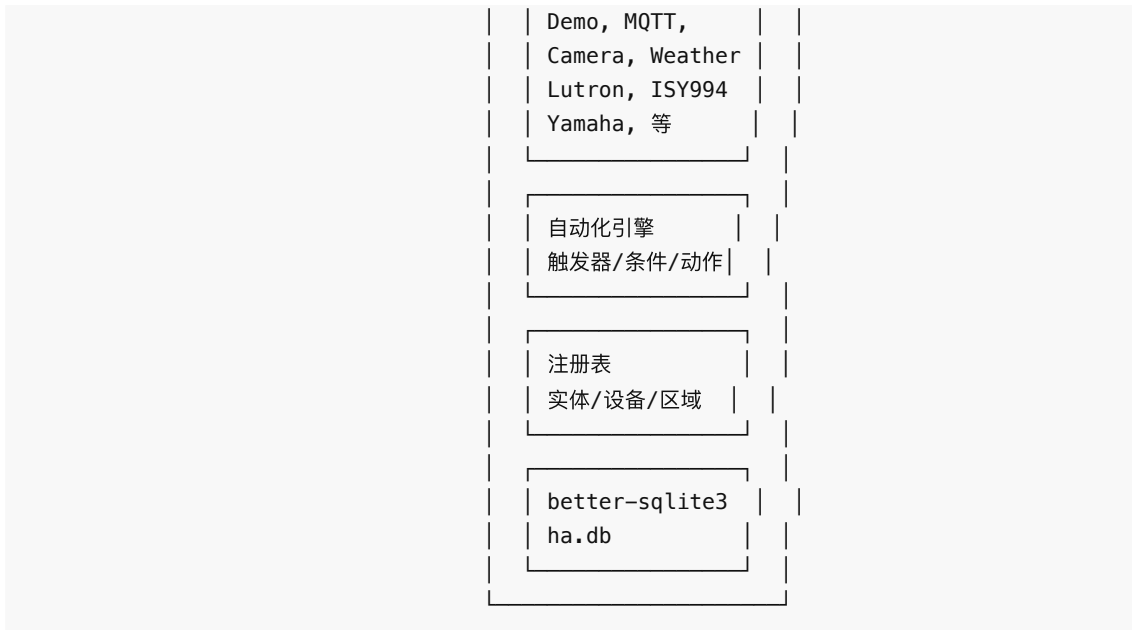
Home Assistant (NestJS)

1. [_____](#)
2. [_____](#)
3. [_____](#)
4. [: _____](#)
5. [: _____\(core\)](#)
6. [: _____\(auth\)](#)
7. [: API _____\(api\)](#)
8. [: WebSocket _____\(websocket\)](#)
9. [: _____\(registry\)](#)
10. [: _____\(integrations\)](#)
11. [: _____\(automations\)](#)
12. [: _____\(database\)](#)
13. [: _____](#)
14. [: _____\(context\)](#)
15. [: _____\(pages\)](#)
16. [: _____\(cards\)](#)
17. [: _____\(components\)](#)
18. [: Three.js 3D _____\(hooks\)](#)
19. [: _____\(utils\)](#)
20. [MCP _____\(mcp-server\)](#)
21. [_____ \(deploy.sh\)](#)
22. [PM2 _____\(ecosystem.config.js\)](#)

1.

Home Assistant	NestJS/TypeScript		NestJS
TypeORM + better-sqlite3	React 18	Three.js	3D
Socket.IO			





2.

```
home_assistant/
├── package.json          # 后端依赖和脚本
├── nest-cli.json         # NestJS CLI 配置
├── tsconfig.json         # TypeScript 配置
├── tsconfig.build.json   # 构建 TypeScript 配置
├── ecosystem.config.js   # PM2 进程管理配置
├── deploy.sh            # 远程部署脚本
├── .env.example          # 环境变量模板
├── .github/workflows/ci.yml # CI 配置
├── README.md            # 项目说明
├──
├── config/              # 运行时配置文件
│   ├── configuration.yaml # 主配置 (HA 风格)
│   ├── configuration.example.yaml # 完整配置示例
│   ├── automations.yaml  # 自动化规则
│   └── floors.json       # 3D 楼层配置
├──
├── src/                 # 后端 TypeScript 源码
│   ├── main.ts          # 应用入口
│   ├── app.module.ts     # 根模块
│   ├── config/          # 配置加载器
│   ├── database/        # 数据库配置
│   ├── common/          # 公共工具 (类型/常量/过滤器/拦截器)
│   ├── core/            # 核心引擎 (事件总线/状态机/服务注册表)
│   ├── auth/            # 认证系统 (JWT/本地策略)
│   ├── api/             # REST API 控制器
│   ├── websocket/       # WebSocket 网关和处理程序
│   ├── registry/        # 实体/设备/区域注册表
│   └── integrations/    # 集成系统 (13 个内置集成)
```

```

├── automations/          # 自动化引擎
├── frontend/            # React 前端源码
│   ├── index.html       # HTML 入口
│   ├── package.json     # 前端依赖
│   ├── vite.config.ts   # Vite 构建配置
│   ├── tsconfig*.json   # TypeScript 配置
│   └── src/
│       ├── main.tsx     # React 入口
│       ├── App.tsx      # 根组件
│       ├── index.css    # 全局样式 (iOS 风格)
│       ├── types.ts     # 类型定义
│       ├── context/     # React Context (状态管理)
│       ├── components/  # UI 组件
│       ├── pages/       # 页面组件 (11 个页面)
│       ├── hooks/       # Three.js 3D 场景 Hooks
│       └── utils/       # 音效/语音工具
├── public/              # 构建输出 (编译后的前端)
│   └── assets/          # JS/CSS 分包
├── data/                # 运行数据
│   └── floors/          # 上传的 .glb 3D 模型
├── mcp-server/          # MCP AI 服务
│   ├── index.js         # MCP 服务器入口
│   ├── package.json     # MCP 依赖
│   └── claude_desktop_config.json # Claude Desktop 配置
├── ha.db                # SQLite 数据库文件
└── dist/                # 编译后的后端

```

3.

3.1 package.json

NestJS :

- **@nestjs/*** — NestJS (core, common, config, jwt, passport, platform-express, platform-socket.io, swagger, typeorm, websockets)
- **typeorm** — ORM
- **better-sqlite3** — SQLite
- **socket.io** — WebSocket
- **passport / passport-jwt / passport-local** —
- **bcrypt** —
- **eventemitter2** —
- **joi / class-validator / class-transformer** —
- **mqtt** — MQTT
- **hls.js** — HLS
- **uuid** — UUID
- **yaml / xml2js** —

:

npm start	(nest start)
npm run start:dev	()
npm run start:prod	node dist/main
npm run build	+
npm test	Jest
npm run migration:run	TypeORM

3.2 tsconfig.json

TypeScript ES2021 CommonJS :

@core/*	src/core/*
@registry/*	src/registry/*
@auth/*	src/auth/*
@common/*	src/common/*
@integrations/*	src/integrations/*
@websocket/*	src/websocket/*
@api/*	src/api/*

3.3 nest-cli.json

NestJS CLI Swagger API

3.4 .env.example

HA_JWT_SECRET HA_HTTP_PORT

3.5 config/configuration.yaml

— Home Assistant :

```
homeassistant:
  name: "My Home"           # 家居名称
  latitude: 39.9042         # 纬度
  longitude: 116.4074       # 经度
  elevation: 43             # 海拔
  unit_system: metric       # 单位制 (metric/imperial)
  time_zone: "Asia/Shanghai" # 时区
  country: "CN"            # 国家
```

```
currency: "CNY"           # 货币

http:
  port: 8123               # 监听端口
  cors_allowed_origins:   # CORS 白名单
    - "http://localhost:5173"

auth:
  jwt_secret: "${HA_JWT_SECRET}" # JWT 密钥（环境变量）
  jwt_expiry: "24h"             # Token 过期时间

database:
  type: "better-sqlite3"      # 数据库类型
  database: "ha.db"           # 数据库文件
  logging: false              # SQL 日志

logger:
  level: "info"               # 日志级别

integrations:
  - domain: demo              # 启用的集成

automation: automations.yaml  # 自动化配置
```

3.6 config/configuration.example.yaml

```
(313 ) :demo yamaha_avr weather light switch sensor camera
mqtt lutron_caseta binary_sensor isy994
```

3.7 config/automations.yaml

```
5 :

• motion_light: 5
• night_dim: 22:00
• alarm_door_open:
• alarm_window_open:
• alarm_disarm:
• high_temp_alert: 35°C
```

3.8 config/floors.json

```
3D :
```

```
[
  {"id":"1","name":"Main Floor", "glb":"floor1.glb"},
  {"id":"2","name":"Upper Floor","glb":"floor2.glb"},
  {"id":"3","name":"Basement", "glb":"floor3.glb"}
]
```

4. :

src/main.ts —

```
: NestJS

:

1. NestFactory.create<NestExpressApplication>(AppModule)
2. :
   o public/ — SPA
   o data/ — (GLB )
3. CORS ( /api ) (ValidationPipe) (HaExceptionFilter)

4. Swagger ( /api/doc )
5. SPA : /api/ /data/ /assets/ index.html
6. 8123
```

src/app.module.ts —

```
:

@Module({
  imports: [
    ConfigModule.forRoot({ load: [loadHaConfig] }), // YAML 配置
    TypeOrmModule.forRootAsync({ useFactory: getTypeOrmConfig }), // 数据库
    CoreModule, // 核心引擎（全局模块）
    RegistryModule, // 注册表
    AuthModule, // 认证
    WebSocketModule, // WebSocket 网关
    ApiModule, // REST API
    IntegrationsModule, // 集成
  ]
})

:

1. homeassistant_start —
2.
3. homeassistant_started —
```

5. : (core)

5.1 src/core/core.module.ts —

```
@Global()
```

5.2 (src/core/event-bus/)

```
: event-bus.service.ts
```

```
: EventEmitter2 /
```

```
其他服务 —fire(event, data)→ 事件总线 —emit→ 订阅者
|
```

EventEmitter2 (通配符支持)

:

- `fire(eventType, data, context?)` —
 - `listen(eventType, handler)` — (*)
 - `listenOnce(eventType, handler)` —
 - `listenMany(eventTypes, handler)` —
 - `listenerCount(eventType)` —
 - `removeAllListeners(eventType?)` —
- (`events.constants.ts`):
- `STATE_CHANGED` —
 - `CALL_SERVICE` —
 - `SERVICE_REGISTERED / SERVICE_REMOVED` — /
 - `ENTITY/DEVICE/AREA_REGISTRY_UPDATED` —
 - `HOMEASSISTANT_START / HOMEASSISTANT_STARTED / HOMEASSISTANT_STOP` —
 - `COMPONENT_LOADED` —
 - `USER_UPDATED` —
- (`ha-event.interface.ts`):
- `HaEvent<T>` —
 - `StateChangedData` —
 - `CallServiceData` —
 - `ComponentLoadedData` —

5.3 (`src/core/state-machine/`)

: `state-machine.service.ts`

:



(`ha-state.interface.ts`):

```
interface HaState {
  entity_id: string // 如 "light.living_room"
  state: string // 如 "on", "off", "23.5"
  attributes: Record // 如 { brightness, friendly_name }
  last_changed: string // 上次状态变化时间戳
  last_updated: string // 上次更新(含属性)时间戳
  context: { id, parent_id, user_id } // 上下文
}
```

```

:
• setState(entityId, state, attributes?, context?) — ( )
• getState(entityId) —
• getStates() —
• hasEntity(entityId) —
• removeEntity(entityId) —
• getHistory(entityId, start, end?) —
• getHistoryMultiple(entityIds, start, end?) —
:
• ( Object.freeze )
• last_changed ; last_updated
•
• states_history
( state.entity.ts ):

```

```

@Entity('states_history')
class StateHistoryEntity {
  @PrimaryGeneratedColumn() id: number
  @Column() entity_id: string
  @Column() state: string
  @Column('simple-json') attributes: Record<string, unknown>
  @Column() last_changed: Date
  @Column({ nullable: true }) context_id: string
  @Column({ nullable: true }) context_user_id: string
  @Column({ nullable: true }) context_parent_id: string
}

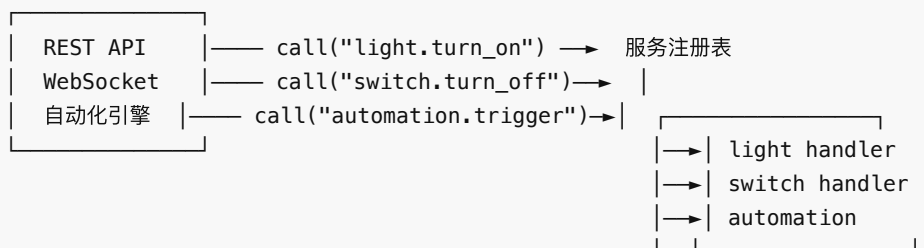
```

5.4 (src/core/service-registry/)

```

: service-registry.service.ts
: domain.service_name

```



```

( ha-service.interface.ts ):

```

```

• ServiceField —
• ServiceTarget — (entity_id, area_id, device_id)
• ServiceCall —
• ServiceHandler — (call: ServiceCall) => Promise<void>

```


- `ServiceDescriptor` —
- `:`
- `register(domain, service, handler, descriptor?)` —
- `call(domain, service, data, target?)` —
- `has(domain, service)` —
- `getDescriptor(domain, service)` —
- `getDomainServices(domain)` —
- `getAllServices()` —
- `remove(domain, service)` —
- `validate(domain, service, data)` —

5.5 (`src/core/context/`)

```

: context.service.ts , ha-context.interface.ts

: ( ID ID ID )

```

6. : (auth)

`src/auth/auth.module.ts`

`PassportModule JwtModule TypeOrmModule`

6.1 `auth.service.ts` —

```

: JWT Token

:

• validateUser(username, password) — (bcrypt )
• login(user) — JWT access_token
• validateToken(token) — JWT + Token
• createUser(username, password, displayName) —
• createLongLivedToken(userId, name) — Token (SHA-256 )
• listLongLivedTokens(userId) — Token
• revokeLongLivedToken(tokenId) — Token
• findUserById(id) —
• decodeToken(token) — JWT

: (admin/admin)

```

6.2 `strategies/jwt.strategy.ts` — JWT

`Authorization: Bearer <token>` JWT

6.3 `strategies/local.strategy.ts` —

/

6.4 `guards/jwt-auth.guard.ts` — JWT

```

: JWT Token (LLT) LLT X-HA-access-token

Authorization: Bearer

```

6.5 guards/local-auth.guard.ts —

Passport

6.6 entities/ —

user.entity.ts :

```
@Entity('users')
class UserEntity {
  @PrimaryGeneratedColumn('uuid') id: string
  @Column({ unique: true }) username: string
  @Column() password_hash: string
  @Column({ nullable: true }) display_name: string
  @Column({ default: true }) is_admin: boolean
  @Column({ default: true }) is_active: boolean
  // 时间戳
}
```

long-lived-token.entity.ts :

```
@Entity('long_lived_tokens')
class LongLivedTokenEntity {
  @PrimaryGeneratedColumn('uuid') id: string
  @Column() user_id: string
  @Column() name: string
  @Column() token_hash: string // SHA-256
  @Column({ nullable: true }) expires_at: Date
  @Column({ nullable: true }) last_used_at: Date
}
```

6.7 decorators/current-user.decorator.ts

@CurrentUser() —

6.8 auth.controller.ts — API

/api/auth/login	POST		JWT
/api/auth/token	POST		OAuth2 Token
/api/auth/long_lived_tokens	GET	JWT	Token
/api/auth/long_lived_tokens	POST	JWT	Token
/api/auth/long_lived_tokens/:id	DELETE	JWT	Token

7. : API (api)

src/api/api.module.ts

API

REST

7.1 health.controller.ts —

GET /api/		{"status":"ok","uptime":秒数}
GET /api/health		{"status":"ok","uptime":秒数}

7.2 states.controller.ts —

/api/states	GET	JWT	
/api/states/:entity_id	GET	JWT	
/api/states/:entity_id	POST	JWT	

POST /api/states/:entity_id :

```
{
  "state": "on",
  "attributes": { "brightness": 128 }
}
```

7.3 services.controller.ts —

/api/services	GET	JWT	
/api/services/:domain/:service	POST	JWT	

7.4 events.controller.ts —

/api/events	GET	JWT	
/api/events/:event_type	POST	JWT	

7.5 history.controller.ts —

/api/history/period/:timestamp	GET	JWT	

: filter_entity_id — , end_time —

7.6 config.controller.ts —

/api/config	GET	JWT	HA
/api/config/integrations-status	GET	JWT	
/api/config/automations	GET	JWT	YAML
/api/config/automations	PUT	JWT	YAML
/api/config/3d-mappings	GET	JWT	3D
/api/config/3d-mappings	PUT	JWT	3D
/api/config/floors	GET	JWT	
/api/config/floors	POST	JWT	
/api/config/floors/:id	DELETE	JWT	
/api/config/apply	POST	JWT	

7.7 registry.controller.ts —

/api/entity_registry	GET	
/api/entity_registry/:entity_id	GET	
/api/entity_registry/:entity_id	PUT	
/api/device_registry	GET	
/api/device_registry/:device_id	GET	
/api/area_registry	GET	
/api/area_registry	POST	
/api/area_registry/:area_id	PUT	
/api/area_registry/:area_id	DELETE	

7.8 backup.controller.ts —

/api/backup	GET	
/api/backup/restore	POST	

7.9 ai.controller.ts — AI

/api/ai/chat	POST	AI (Claude+)

: { "prompt": "关掉客厅灯" }

AI call_service

7.10 glb.controller.ts — 3D

/api/glb/upload/:floorId	POST	.glb 3D

7.11 camera.controller.ts —

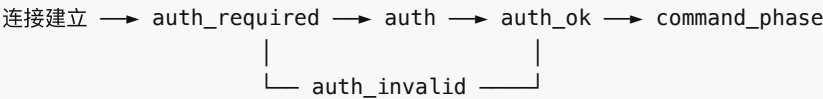
/api/camera/hls/:entityId/index.m3u8	GET	HLS
/api/camera/hls/:entityId/:segment	GET	HLS TS

8. : WebSocket (websocket)

src/websocket/websocket.gateway.ts —

: Home Assistant WebSocket /api/websocket

:



:

```
// 请求
{ "id": 1, "type": "get_states", ... }

// 成功响应
{ "id": 1, "type": "result", "success": true, "result": [...] }

// 事件推送
{ "type": "event", "event": { "event_type": "state_changed", "data": {...} } }
```

8.1 ws-session.service.ts —

WebSocket :

- socketId — ID
- socket — Socket
- phase — (auth_required , auth , command)

- `authenticated` —
- `user` —
- `subscriptions` —

8.2

`IWsHandler`

`ping.handler.ts`

- `:` `ping`
- `:` `pong`

`get-states.handler.ts`

- `:` `get_states`
- `:`

`get-services.handler.ts`

- `:` `get_services`
- `:`

`get-config.handler.ts`

- `:` `get_config`
- `:` `HA`

`call-service.handler.ts`

- `:` `call_service`
- `:` `domain` , `service` , `service_data` , `target`
- `:`

`subscribe-events.handler.ts`

- `:` `subscribe_events`
- `:` `event_type` ()
- `:` `WebSocket`

`unsubscribe-events.handler.ts`

- `:` `unsubscribe_events`
- `:` `subscription`
- `:`

8.3

`ws-session.interface.ts` : `WsSession` `ws-handler.interface.ts` :

9. : (registry)

9.1 `registry.module.ts`

`TypeORM`

9.2 (`entity-registry/`)

(`entity.entity.ts`):

```

@Entity('entity_registry')
class EntityRegistryEntity {
    @PrimaryColumn() entity_id: string // PK, 如 "light.living_room"
    @Column({ nullable: true }) platform: string // 集成平台
    @Column({ nullable: true }) unique_id: string // 唯一 ID
    @Column({ nullable: true }) name: string // 自定义名称
    @Column({ nullable: true }) original_name: string
    @Column({ nullable: true }) icon: string
    @Column({ nullable: true }) device_id: string
    @Column({ nullable: true }) area_id: string
    @Column({ default: false }) disabled: boolean
    @Column({ nullable: true }) disabled_by: string
    @Column({ nullable: true }) unit_of_measurement: string
    @Column({ nullable: true }) device_class: string
}

```

(entity-registry.service.ts):

- registerEntity() —
- updateEntity() — ()
- removeEntity() —
- getEntity() / getEntities() —
- getEntitiesByPlatform() / getEntitiesByArea() / getEntitiesByDevice() —

: + SQLite

9.3 (device-registry/)

(device.entity.ts):

```

@Entity('device_registry')
class DeviceEntity {
    @PrimaryColumn() device_id: string // PK
    @Column({ nullable: true }) name: string
    @Column({ nullable: true }) manufacturer: string
    @Column({ nullable: true }) model: string
    @Column({ nullable: true }) sw_version: string
    @Column({ nullable: true }) hw_version: string
    @Column({ nullable: true }) area_id: string
    @Column('simple-json', { nullable: true }) identifiers_json: string[]
    @Column({ nullable: true }) integration: string
    @Column({ nullable: true }) via_device_id: string
    @Column({ nullable: true }) configuration_url: string
}

```

(device-registry.service.ts):

- registerDevice() — (identifiers)
- findByIdentifiers() —
- getDevicesInArea() —

9.4 (area-registry/)

(area.entity.ts):

```
@Entity('area_registry')
class AreaEntity {
  @PrimaryColumn()      area_id: string      // PK, 如 "living_room"
  @Column()              name: string
  @Column('simple-json', { nullable: true }) aliases_json: string[]
  @Column({ nullable: true }) picture: string
}
```

(area-registry.service.ts):

- CRUD
- area_registry_updated

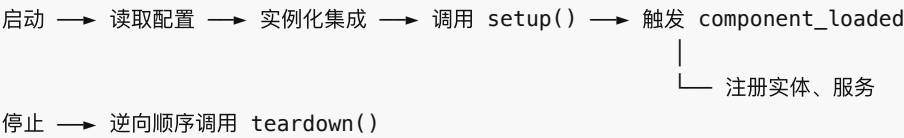
10. : (integrations)

integrations.module.ts

RegistryModule AutomationModule

10.1 integration-loader.service.ts —

:



:

- 13
- /
-
-

10.2 integration.interface.ts —

```
interface HaIntegration {
  setup(): Promise<void>      // 初始化 (注册实体、服务)
  teardown(): Promise<void>  // 清理
}

interface IntegrationManifest {
  domain: string
  name: string
  version?: string
}
```



```
dependencies?: string[]
}
```

10.3

10.3.1 Demo (demo/)

```
: demo.integration.ts
```

```
: 29
```

```
:
```

- 6 (light.living_room, light.bedroom, light.kitchen)
- 3 (switch.fan, switch.tv, switch.alarm_siren)
- 11 (binary_sensor.front_door, binary_sensor.back_door)
- 4 (sensor.temperature, sensor.humidity)
- 2 (media_player.living_room_speaker, media_player.bedroom_speaker)
- 3

```
: turn_on, turn_off, toggle ( light/switch/media_player )
```

10.3.2 Light (light/)

```
: light.integration.ts
```

```
: light.turn_on/turn_off/toggle
```

```
turn_on :
```

- brightness (0-255) —
- brightness_pct (0-100) —
- color_temp —
- rgb_color — RGB
- effect —
- transition —

```
3D : glb_mesh glb_floor glb_pos 3D
```

10.3.3 Switch (switch/)

```
: switch.integration.ts
```

```
: switch.turn_on/turn_off/toggle
```

10.3.4 Sensor (sensor/)

```
: sensor.integration.ts
```

```
: 30 :
```

	18-35°C	°C
	30-80%	%

	980-1040	hPa
	0-100000	lux
CO2	350-2000	ppm

10.3.5 Binary Sensor (binary-sensor/)

: binary-sensor.integration.ts

: () device_class : motion door window garage_door curtain blind
presence smoke moisture gas

3D : glb_floor glb_pos glb_mesh 3D

10.3.6 MQTT (mqtt/)

: mqtt.integration.ts , mqtt-client.service.ts

: MQTT Home Assistant

:

- mqtt.integration.ts —
- mqtt-client.service.ts — MQTT (/ / / /)
- mqtt.constants.ts —

: light, switch, sensor, binary_sensor

:

-
- (Broker)
-
- ON/OFF/TOGGLE

(configuration.yaml):

```
- domain: mqtt
  broker: localhost
  port: 1883
  discovery_prefix: homeassistant
```

10.3.7 Camera (camera/)

: camera.integration.ts , camera-stream.service.ts , camera.controller.ts

: HLS

:

1. RTSP (SD/HD)
2. camera
3. FFmpeg (RTSP → HLS)
4. 30
5. 5 ()

FFmpeg :

- SD: 640x360, 15fps, 500kbps
- HD: 1280x720, 15fps, 1000kbps
- : AAC, 32kbps

HLS : `/api/camera/hls/:entityId/{index.m3u8, *.ts}`

10.3.8 Weather (`weather/`)

: `weather.integration.ts` , `weather.constants.ts`

: Open-Meteo API (API)

:

- 1.
2. Nominatim API
3. Open-Meteo API
4. 10

:

- +
-
- + (kmh/mpH/ms)
-
- WMO → HA (sunny, cloudy, rainy)
- : (°C) (°F) `unit_system`

10.3.9 Yamaha AVR (`yamaha-avr/`)

: `yamaha-avr.integration.ts` , `yamaha-avr-client.service.ts`

: Yamaha AV (HTTP+XML YNCA)

: YNCA Yamaha

:

- Zone `media_player`
- Zone : Main_Zone, Zone_2, Zone_3

:

- turn_on / turn_off —
- volume_up / volume_down —
- volume_set — (0-1)
- mute_volume —
- select_source —

:

- HTTP+XML
- Mock ()
- dB ⇔ 0-1

10.3.10 Lutron Caseta (`lutron-caseta/`)

```
: lutron-caseta.integration.ts , lutron-caseta-client.service.ts
```

```
: Lutron Caseta ( Telnet )
```

```
:
```

- dimmer → light ()
- switch → switch ()
- fan → switch ()

```
:
```

1. Telnet Lutron Smart Bridge
- 2.
- 3.
4. ()
- 5.

```
: light.turn_on/turn_off, switch.turn_on/turn_off
```

```
: lutron / integration
```

10.3.11 ISY994 / Insteon (isy994/)

```
: isy994.integration.ts , isy994-client.service.ts , isy994-subscription.service.ts ,  
isy994-utils.ts
```

```
: ISY994 ( Insteon Z-Wave )
```

```
:
```

- isy994.integration.ts —
- isy994-client.service.ts — HTTP+XML REST
- isy994-subscription.service.ts — WebSocket
- isy994.utils.ts — ()
- isy994.constants.ts —

```
:
```

1. REST API
2. domain:
 - (1) → light
 - (2) → switch
 - (7) → binary_sensor
 - (11) →

- 3.

```
:
```

- DON — ()
- DOF —
- DFON —
- DFOF —
- BRT —
- DIM —

```
: light.turn_on/turn_off, switch.turn_on/turn_off, binary_sensor
```

- WebSocket ()
- Mock
- 60

```
: envisalink.integration.ts , envisalink.constants.ts
```

$$\vdots$$

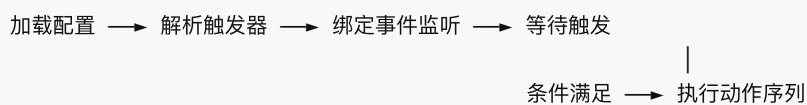
- Partition → alarm_control_panel (/)
- Zone → binary_sensor ()

- `arm_stay` —
- `arm_away` —
- `disarm` — ()

•

automation.module.ts

11.1 automation-engine.service.ts —

$$\vdots$$

$$\vdots$$

```
- id: "motion_light"
  alias: "Turn on living room light when motion detected"
  trigger:
    - platform: state
      entity_id: binary_sensor.motion
      to: "on"
  condition:
    - condition: state
      entity_id: light.living_room
      state: "off"
```

```
action:
  - service: light.turn_on
    target:
      entity_id: light.living_room
mode: single
```

- :
- single —
 - parallel —
 - restart — ()
- :
- (A B A)
 -

11.2 trigger-evaluator.service.ts —

:

state	entity_id, from, to, for?	
numeric_state	entity_id, above?, below?, for?	(>= / <=)
time	hours, minutes, seconds	(HH:MM)
event	event_type, event_data?	

- for: :
-

11.3 condition-evaluator.service.ts —

:

state	entity_id, state	
numeric_state	entity_id, above?, below?	
time	after?, before?, weekday?	()

11.4 action-executor.service.ts —

:

:

service	service, target, data	

delay	seconds, minutes, hours	
condition	condition, then?, else?	

:

- (10)

11.5 automation.integration.ts —

Home Assistant :

- automation.trigger —
- automation.turn_on —
- automation.turn_off —

config/automations.yaml

12. : (database)

src/database/typeorm.config.ts

: TypeORM

```
export function getTypeOrmConfig(): TypeOrmModuleOptions {
  return {
    type: 'better-sqlite3',
    database: 'ha.db',           // 配置文件路径
    entities: [__dirname + '/../**/*.entity{.ts,.js}'],
    synchronize: true,          // 自动同步 schema
    logging: false,
  }
}
```

:

users	UserEntity	
long_lived_tokens	LongLivedTokenEntity	API Token
states_history	StateHistoryEntity	
entity_registry	EntityRegistryEntity	
device_registry	DeviceEntity	
area_registry	AreaEntity	

: synchronize: true ()

13. :

13.1 frontend/package.json

:

- **react 18** — UI
- **three.js 0.183** — 3D
- **socket.io-client** — WebSocket
- **hls.js** — HLS

: Vite 5 + TypeScript 5

13.2 frontend/vite.config.ts —

:

- : ../public ()
- : three.js react socket.io-client hls.js chunk
- : 5173 /api localhost:8123 (WebSocket)

13.3 frontend/index.html — HTML

- PWA : manifest.json + Service Worker
- iOS : apple-mobile-web-app-capable , viewport-fit=cover
- : #1c1c1e ()

13.4 src/main.tsx — React

```
createRoot(document.getElementById('root')!).render(  
  <StrictMode><App /></StrictMode>  
)
```

13.5 src/types.ts —

:

```
interface MappingEntry {  
  entity: string  
  behavior: string  
}  
type Mappings = Record<string, string | MappingEntry> // mesh → entity 映射  
type BehaviorMap = Record<string, string> // mesh → behavior 映射  
type FloorId = 1 | 2 | 3 // 楼层 ID
```

3D : GlbLightCfg , SphereLightCfg , SensorCfg , SensorGlbCfg —

3D

13.6 src/index.css —

: iOS (SF Pro)

CSS :

- --bg / --bg2 —
- --surface / --surface2 / --surface3 —

- `--text / --text2 / --text3` —
 - `--blue / --green / --red / --orange` — iOS
 - `--radius / --radius-lg` —
 - `--tab-h / --nav-h` —
 - `--safe-bottom` —
- `/` — `html.light`

14. : (context)

14.1 HaContext.tsx — HA

: React Context HA

:

```
interface HaCtx {
  token: string | null           // JWT Token
  login(username, password)     // 登录
  logout()                       // 登出
  wsConnected: boolean          // WebSocket 连接状态
  health: HaHealth | null       // 服务器健康状态
  states: Map<string, HaState>   // 所有实体状态
  callService(domain, service, data?, entityId?) // 调用服务
  setEntityState(entityId, state, attributes?) // 设置状态
}
```

:

1. : `localStorage` Token
2. : `POST /api/auth/login` → Token → WebSocket
3. **WebSocket** :
 - `Socket.IO` `/api/websocket`
 - : `auth_required` → `auth` → `auth_ok`
 - `get_states` + `subscribe_events`
 - `state_changed` Map
4. : `Map<string, HaState>` — ID →
5. : Token WebSocket

14.2 ToastContext.tsx —

: toast

```
interface ToastCtx {
  toast(message: string, type?: 'success' | 'error' | 'info')
}
```

Toast

3

15. : (pages)

App.tsx

Suspense

TabBar

15.1 App.tsx —

:

- /floorplan /3d → 3D ()
- → LoginPage
- → TabBar +

AI : ◆ AI (createPortal document.body)

15.2 LoginPage.tsx —

- admin/admin
- POST /api/auth/login
- iOS

15.3 DashboardPage.tsx —

:

:

- 1.
- 2.
3. + ()
4. :
5. (pinned)
- 6.

:

- ()
- WebSocket ()
-

15.4 EntitiesPage.tsx —

:

:

-
-
- (DashboardPage)
- iOS
-

15.5 FloorPlanPage.tsx — 3D

: Three.js 3D

:

- (+ 1/2/3)
- GLB 3D (Blender/SketchUp)
- 3D

- (+)
- : 3D
- :
- (URL ?filter=light,door)
- : / / (3)
- /
- (EN/ /فارسی)
- ()
- / (20)

:

1. Three.js (useThreeScene)
2. 3D
3. GLB (useSceneContent)
4. (glb_mesh/glb_floor/glb_pos)
5. (useSceneClick)
6. ()

15.6 HistoryPage.tsx —

:

:

1. + (1h/6h/24h/7d)
2. /api/history/period/:timestamp
3. Canvas 2D
4. Y X

15.7 EventsPage.tsx —

: WebSocket

:

1. WebSocket state_changed
2. (500)
3. /

15.8 AutomationsPage.tsx —

:

:

- : / /
- YAML : automations.yaml

15.9 AreasPage.tsx —

(area_registry)

15.10 IntegrationsPage.tsx —

:

: Demo, MQTT, EnvisaLink, ISY994, Lutron Caseta

-
- /
- / (MQTT EnvisaLink)
-
- : ✓ Loaded / ✕ Failed / ⇄ Pending
- : / / GLB

15.11 SettingsPage.tsx —

- (/)
- WebSocket
-
-
-
-

15.12 BackupPage.tsx —

- SQLite
-

16. : (cards)

HaState props

16.1 LightCard.tsx —

- (On/Off/)
- iOS
- ()

16.2 SwitchCard.tsx —

- +
- iOS toggle

16.3 SensorCard.tsx —

- (sensor.*): +
 - (binary_sensor.*): On/Off Open/Closed Detected/Clear
- :

device_class		On	Off
motion		Detected	Clear
door		Open	Closed
window		Open	Closed
smoke		Detected	Clear

moisture		Wet	Dry
temperature		—	—

16.4 WeatherCard.tsx —

- ()
- (🌞 ☁️ 🌧️ ❄️)
- +
-
-

16.5 CameraCard.tsx —

- : +
- (HLS.js)
- modal
- (SD/HD)
- /
- LIVE

16.6 MediaPlayerCard.tsx —

- / /
-
-
- : turn_on/off, volume_up/down/set, mute, select_source

17. : (components)

17.1 TabBar.tsx —

iOS Tab 9 : Home, Devices, 3D, History, Events, Auto, Areas, Integ, More
(backdrop-filter: blur(20px))

17.2 AiChatPanel.tsx — AI

- : AI
- ()
- → POST /api/ai/chat
- AI

17.3 EditPanel.tsx — 3D

- : 3D (mesh) HA
- GLB
- ✓
- (DevicePicker)
- (BehaviorSelect): light, door_r, door_s, window, curtain, garage_door, media_player, switch
-

•

17.4 DevicePicker.tsx —

-
-
- guessBehavior() — entity_id
- BrightnessSlider — (+/- +)
- BehaviorSelect — 3D

18. : Three.js 3D (hooks)

18.1 useThreeScene.ts — Three.js

- : Three.js
- :
- : WebGL, antialias, PCFSoftShadowMap, ACESFilmicToneMapping
- : #2a2a4e
- : AmbientLight + HemisphereLight + DirectionalLight (with shadow)
- : PerspectiveCamera 48° FOV
- : OrbitControls ()
- : SceneHandle — renderer, scene, camera, controls

18.2 useSceneContent.ts — 3D

- : 3D
- :

GLB	3D mesh	+ / + PointLight
Sphere	+ + PointLight	+
Sensor		(/) +
Sensor GLB	3D mesh / /	/
Media Player	3D mesh +	

- :
- door_r — Z 75°
- door_s — Z 0.6m
- garage_door / curtain — (clip plane)
- light —
- switch —
- media_player — + (9)
- : URL ?filter=light,door

18.3 useSceneClick.ts — 3D

```
: 3D

:

1. PointerDown/Move/Up
2. (5px )
3. Raycaster
4. userData.entityId userData.meshName
5. onHit
```

19. : (utils)

sounds.ts —

```
: Web Audio API +

:
```

playDing()		1000Hz
playLightToggle(on)		1800→1200Hz (on) / 1200→800Hz (off)
playDoorToggle(open)		+ +
playGarageToggle(open)		+
playCurtainToggle(open)		+
playMediaToggle(on)		440→880Hz (on)
playSwitchToggle(on)		1500Hz (on) / 1000Hz (off)

```
: startMusic() / stopMusic() — /music.wav ( )

:

• Web Speech API ( SpeechSynthesis )
• 3 : (en-US) (zh-CN) (fa-IR)
• ( / / )
•
• 0.9, 1.1
```

20. MCP (mcp-server)

20.1 index.js — MCP Server

```
: Model Context Protocol (MCP) AI ( Claude Desktop) Home Assistant

:
```

get_states	
get_state	
call_service	
get_services	
get_health	
get_history	

20.2 `claude_desktop_config.json`

Claude Desktop

20.3 `package.json`

: `@modelcontextprotocol/sdk` + `zod`

21. `(deploy.sh)`

```
:
:
1. ( npm run build:frontend )
2. ( npm run build:backend )
3. Rsync ( node_modules, .git, src )
4. PM2
5.
: ./deploy.sh [user@host] [ssh-port]
```

22. PM2 `(ecosystem.config.js)`

```
:
:
• : home-assistant
• : dist/main.js
• : 10
• : 500MB
• :
```

:

3D →

1. 前端 FloorPlanPage
 - ├─ useSceneClick 检测点击
 - ├─ 提取 entityId = "light.living_room"
 - ├─ 调用 haSetState("light.living_room", "off")
 - └─ POST /api/states/light.living_room { state: "off" }
2. 后端 StatesController
 - ├─ JwtAuthGuard 验证 Token
 - └─ 调用 stateMachine.setState("light.living_room", "off")
3. 状态机
 - ├─ 更新内存中的状态 Map
 - ├─ 计算 last_changed (状态实际变化时)
 - ├─ 异步写入 states_history 表
 - └─ 触发 STATE_CHANGED 事件
4. 事件总线
 - ├─ 通知 EventBusService 所有订阅者
 - ├─ WebSocket 网关收到事件
 - └─ 通过 WebSocket 推送给所有连接的前端
5. 前端 HaContext
 - ├─ 收到 state_changed 事件
 - ├─ 更新全局 states Map
 - └─ 触发 React 重新渲染
6. DashboardPage / FloorPlanPage
 - ├─ LightCard 显示 "Off"
 - └─ 3D 场景中灯泡熄灭, 发光消失

1. 集成传感器更新状态
 - ├─ sensor.integration 定时器触发
 - ├─ stateMachine.setState("sensor.temperature", "36.5")
 - └─ STATE_CHANGED 事件触发
2. 自动化引擎
 - ├─ TriggerEvaluator 检测 state 触发器
 - ├─ numeric_state: above 35°C → 条件满足
 - ├─ ConditionEvaluator 检查条件
 - ├─ ActionExecutor 执行动作序列
 - └─ callService("switch.turn_on", { entity_id: "switch.fan" })
3. 服务注册表
 - ├─ 调用 switch.integration 的 handler
 - ├─ stateMachine.setState("switch.fan", "on")
 - ├─ STATE_CHANGED 事件触发
 - └─ WebSocket 推送到前端
4. 前端收到更新

- └─ DashboardPage 显示风扇"On"
- └─ (如果有 3D 映射) FloorPlanPage 更新风扇状态

~100

15,000+ TypeScript/TSX